

04

API & System Integration

Connect your applications, services and external systems through reliable API integrations.

Project Overview

A client was operating a CRM, an accounting tool and an internal Laravel application that did not talk to each other. The same records were being entered up to three times. We built an integration layer that keeps all three systems in sync.

The Challenge

- The same customer and invoice data was entered manually into three systems.
- Data drifted out of sync, so reports from different systems disagreed.
- Third-party APIs had inconsistent formats, rate limits and occasional downtime.
- Failed transfers were silent, so nobody knew when data was missing.

Our Solution

- Designed a Laravel-based integration service exposing a clean, documented REST API.
- Mapped and normalised data formats between systems with a shared transformation layer.
- Implemented queued jobs with retry, backoff and idempotency so temporary failures self-heal.
- Added request/response logging and alerting so every failed sync is visible.
- Secured endpoints with token authentication, validation and rate limiting.
- Version-controlled the integration in Git with a staged testing environment before release.

Technologies Used

Frontend	HTML5, CSS3, JavaScript, Bootstrap, Vue.js
Backend	PHP, Laravel, REST API
Database	MySQL
Dev & Collaboration	Git, GitHub, VS Code, XAMPP

Results

Metric	Before	After
Duplicate data entry	3 systems	1 entry point
Sync delay between systems	Up to 24 hours	Near real time
Silent sync failures	Undetected	Logged and alerted
Monthly reconciliation effort	approx. 10 hours	approx. 1 hour

Key Takeaway

A good integration is judged by how it behaves when something fails. Retries, idempotency and visible logging matter more than the happy path, because external systems will always go down eventually.

Note: Figures shown are representative project outcomes and can be replaced with your own verified client data before sharing.